19 At the start of day n, the number of cells being used in an experiment is T_n

At the start of the next day, the number of cells being used in the experiment is T_{n+1} where

$$T_{n+1} = kT_n \text{ and } k \text{ is a positive constant.}$$

Given that

$T_1 = 250\,000$

$T_3 = 490\,000$

$T_7 : T_4 = m : 1$

find the value of m.
Give your answer correct to 3 significant figures.

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(Total for Question 19 is 5 marks)